Supplementary Material for "Certifying AI-Agent Leaderboard Records: Selective Tests, Online Guarantees and the Winner's Curse"

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This supplement holds the tables and the figure that the main text points to but does not need on the page: the record test's false-rejection rate by position in the chain and its power (Section 5.1 of the paper), the ranking lottery (Section 5.4), the full winner's-curse validation tables for SWE-bench Verified, Terminal-Bench 2 and LiveBench (Section 5.3), and the record ladders of five leaderboards (Section 6). Every number here is generated by the same script as the main text (`make_numbers.py` in the repository, https://github.com/jopoku16/honestboard), from the same result files.

1 The record test by position in the chain, and its power

Tasks	Step in the chain	Steps	selective	two-sided	one-sided
300	first (holder is the first submission)	576	0.7%	0.7%	3.1%
300	second	501	0.2%	0.2%	0.8%
300	third and later	621	0.0%	0.0%	0.2%
500	first (holder is the first submission)	575	1.0%	1.0%	4.0%
500	second	505	0.6%	0.6%	1.0%
500	third and later	625	0.0%	0.0%	0.2%
2,000	first (holder is the first submission)	578	2.1%	2.1%	4.8%
2,000	second	506	0.4%	0.4%	1.6%
2,000	third and later	729	0.0%	0.0%	0.1%

Table S1: Global null: how often the test wrongly rejects, by the step's position in the record chain (fitted Verified model). The first step compares against the first submission, which is only mildly lucky; later steps compare against the best system so far.

Tasks	1 pt	2 pts	3 pts	5 pts	Power at the formula's MDD
300	0.05	0.13	0.27	0.67	0.77 (MDD 5.5 pts)
500	0.07	0.21	0.46	0.90	0.78 (MDD 4.2 pts)
2,000	0.23	0.72	0.98	1.00	0.77 (MDD 2.1 pts)

Table S2: How often the selective paired test catches a true gap of 1, 2, 3 and 5 points, and a gap equal to the minimum detectable difference given by the formula in Section 3 of the paper. Fitted Verified model, baseline score 77%.

2 The ranking lottery

Tasks per benchmark	Winner is a truly best system (share of runs)	Set covers a truly best system when the winner is not	Median set size
100	0.39	1.00	34
250	0.65	1.00	19
500	0.70	0.99	12

Table S3: Known-truth simulation of ranking on SWE-bench Verified: how often the observed winner really is one of the best systems, and whether the set of possible best systems contains a truly best system when the winner is not one.

3 Full winner's-curse validation tables

Setting	Estimator	Mean error (SE)	Median error	RMSE	Coverage	Length	Defined	Mean, common	Median, common
Known truth, 100 tasks	reported	+3.57 (0.19)	+3.80	5.25	0.83	15.1	100%	+3.63	+3.80
	bootstrap	+1.07 (0.21)	+1.00	4.26	0.88	14.0	100%	+1.23	+1.07
	iterated bootstrap	+0.65 (0.21)	+0.62	4.31	0.90	14.0	100%	+0.86	+0.81
	conditional	+2.51 (0.24)	+2.58	4.91	0.91	19.4	76%	+2.51	+2.58
	hybrid (AKM)	+2.51 (0.24)	+2.58	4.91	0.92	19.3	76%	+2.51	+2.58
	zoom interval (ZF)	+3.57 (0.19)	+3.80	5.25	0.96	24.9	100%	+3.63	+3.80
Known truth, 250 tasks	reported	+1.72 (0.12)	+1.60	2.92	0.89	9.9	100%	+1.84	+2.00
	bootstrap	+0.32 (0.13)	+0.26	2.54	0.93	9.2	100%	+0.48	+0.39
	iterated bootstrap	+0.13 (0.13)	+0.12	2.60	0.91	9.2	100%	+0.31	+0.26
	conditional	+0.80 (0.15)	+0.92	2.91	0.94	13.2	86%	+0.80	+0.92
	hybrid (AKM)	+0.80 (0.15)	+0.92	2.91	0.94	13.3	86%	+0.80	+0.92
	zoom interval (ZF)	+1.72 (0.12)	+1.60	2.92	0.99	15.5	100%	+1.84	+2.00
Known truth, 500 tasks	reported	+0.92 (0.08)	+1.00	1.85	0.94	7.0	100%	+0.95	+1.00
	bootstrap	+0.08 (0.09)	+0.04	1.71	0.95	6.6	100%	+0.12	+0.07
	iterated bootstrap	−0.02 (0.09)	−0.11	1.77	0.94	6.6	100%	+0.03	−0.09
	conditional	+0.05 (0.11)	+0.23	2.10	0.98	9.4	90%	+0.05	+0.23
	hybrid (AKM)	+0.05 (0.11)	+0.23	2.10	0.98	9.5	90%	+0.05	+0.23
	zoom interval (ZF)	+0.92 (0.08)	+1.00	1.85	0.99	10.8	100%	+0.95	+1.00
Equal leaders, 250 tasks	reported	+3.03 (0.11)	+2.75	3.75	0.81	10.2	100%	+3.13	+2.75
	bootstrap	+0.86 (0.12)	+0.65	2.56	0.92	9.2	100%	+1.00	+0.78
	iterated bootstrap	+0.43 (0.13)	+0.24	2.57	0.93	9.2	100%	+0.60	+0.41
	conditional	+1.20 (0.18)	+1.21	3.32	0.96	17.9	78%	+1.20	+1.21
	hybrid (AKM)	+1.20 (0.18)	+1.21	3.32	0.96	14.8	78%	+1.20	+1.21
	zoom interval (ZF)	+3.03 (0.11)	+2.75	3.75	0.97	16.5	100%	+3.13	+2.75
Equal leaders, 500 tasks	reported	+2.19 (0.08)	+2.15	2.70	0.83	7.3	100%	+2.23	+2.35
	bootstrap	+0.65 (0.08)	+0.67	1.81	0.93	6.6	100%	+0.70	+0.71
	iterated bootstrap	+0.35 (0.09)	+0.32	1.80	0.93	6.6	100%	+0.41	+0.39
	conditional	+0.30 (0.14)	+0.57	2.60	0.97	15.0	86%	+0.30	+0.57
	hybrid (AKM)	+0.30 (0.14)	+0.57	2.60	0.97	10.6	86%	+0.30	+0.57
	zoom interval (ZF)	+2.19 (0.08)	+2.15	2.70	0.99	11.6	100%	+2.23	+2.35
Held-out tasks	reported	+2.04 (0.16)	+2.40	4.06	–	9.9	100%	+2.14	+2.40
	bootstrap	+0.52 (0.16)	+0.63	3.68	–	9.2	100%	+0.66	+0.65
	iterated bootstrap	+0.30 (0.17)	+0.41	3.71	–	9.2	100%	+0.45	+0.42
	conditional	+0.72 (0.20)	+0.94	4.12	–	15.7	82%	+0.72	+0.94
	hybrid (AKM)	+0.72 (0.20)	+0.94	4.12	–	14.2	82%	+0.72	+0.94
	zoom interval (ZF)	+2.04 (0.16)	+2.40	4.06	–	15.7	100%	+2.14	+2.40
Held-out repositories (cluster, shift-adjusted)	reported	+3.48 (0.17)	+2.98	5.21	–	9.9	100%	+3.90	+3.70
	bootstrap	+1.81 (0.18)	+1.59	4.34	–	15.6	100%	+2.22	+2.13
	iterated bootstrap	+1.85 (0.18)	+1.97	4.48	–	15.6	100%	+2.24	+2.37
	conditional	+3.38 (0.24)	+3.71	5.90	–	13.1	82%	+3.38	+3.71
	hybrid (AKM)	+3.43 (0.23)	+3.71	5.82	–	12.2	82%	+3.43	+3.71
	zoom interval (ZF)	+3.48 (0.17)	+2.98	5.21	–	13.5	100%	+3.90	+3.70
Held-out repositories (cluster, unadjusted)	reported	+3.82 (0.26)	+3.05	6.96	–	9.9	100%	+3.68	+2.69
	bootstrap	+2.14 (0.26)	+2.29	6.21	–	15.6	100%	+2.00	+1.67
	iterated bootstrap	+2.18 (0.27)	+2.22	6.46	–	15.6	100%	+2.03	+1.58
	conditional	+3.17 (0.31)	+2.01	7.12	–	13.1	82%	+3.17	+2.01
	hybrid (AKM)	+3.22 (0.31)	+2.03	7.08	–	12.2	82%	+3.22	+2.03
	zoom interval (ZF)	+3.82 (0.26)	+3.05	6.96	–	13.5	100%	+3.68	+2.69
Held-out repositories (iid, shift-adjusted)	reported	+3.51 (0.17)	+3.04	5.22	–	9.7	100%	+3.90	+3.75
	bootstrap	+2.13 (0.17)	+1.48	4.45	–	9.1	100%	+2.57	+2.28
	iterated bootstrap	+1.98 (0.18)	+1.30	4.42	–	9.1	100%	+2.45	+2.04
	conditional	+2.85 (0.22)	+2.23	5.21	–	13.4	82%	+2.85	+2.23
	hybrid (AKM)	+2.85 (0.22)	+2.23	5.21	–	13.0	82%	+2.85	+2.23
	zoom interval (ZF)	+3.51 (0.17)	+3.04	5.22	–	15.2	100%	+3.90	+3.75

Table S4: All validation settings on SWE-bench Verified. Error is estimate minus target in percentage points; positive means overstated. Coverage is shown only where the truth is known; length is the median length of the 95% interval in points. The held-out-repository rows show the repository-resampling corrections against the shift-adjusted and the unadjusted target, and the task-resampling (iid) corrections against the shift-adjusted target. The last two columns use the repetitions in which all six estimators are defined.

Setting	Estimator	Mean error (SE)	Median error	RMSE	Coverage	Length	Defined	Mean, common	Median, common
Terminal-Bench 2: known truth, 89 tasks	reported	+1.19 (0.11)	+1.12	2.90	0.85	9.2	100%	+1.20	+1.12
	bootstrap	−0.06 (0.12)	−0.03	2.97	0.85	9.2	100%	−0.01	+0.01
	iterated bootstrap	−0.10 (0.13)	−0.07	3.11	0.83	9.2	100%	−0.03	−0.03
	conditional	−0.18 (0.17)	+0.44	4.15	0.91	10.5	96%	−0.18	+0.44
	hybrid (AKM)	−0.13 (0.15)	+0.43	3.73	0.91	10.7	97%	+0.01	+0.44
	zoom interval (ZF)	+1.19 (0.11)	+1.12	2.90	0.95	14.1	100%	+1.20	+1.12
Terminal-Bench 2: equal leaders, 89 tasks	reported	+2.66 (0.10)	+2.47	3.65	0.82	10.8	100%	+2.73	+2.69
	bootstrap	+0.83 (0.11)	+0.60	2.79	0.90	10.0	100%	+0.92	+0.76
	iterated bootstrap	+0.44 (0.11)	+0.24	2.79	0.90	10.0	100%	+0.55	+0.34
	conditional	+0.50 (0.16)	+0.89	3.80	0.94	18.2	88%	+0.50	+0.89
	hybrid (AKM)	+0.53 (0.16)	+0.89	3.75	0.94	14.5	88%	+0.53	+0.89
	zoom interval (ZF)	+2.66 (0.10)	+2.47	3.65	0.95	16.0	100%	+2.73	+2.69
Terminal-Bench 2: held-out tasks	reported	+1.98 (0.23)	+1.59	5.98	–	13.3	100%	+1.91	+1.12
	bootstrap	−0.71 (0.25)	−0.90	6.21	–	13.7	100%	−0.73	−0.94
	iterated bootstrap	−1.15 (0.26)	−1.21	6.52	–	13.7	100%	−1.15	−1.17
	conditional	−0.70 (0.32)	+0.12	7.72	–	17.6	94%	−0.70	+0.12
	hybrid (AKM)	−0.48 (0.30)	+0.07	7.14	–	16.7	95%	−0.39	+0.12
	zoom interval (ZF)	+1.98 (0.23)	+1.59	5.98	–	20.8	100%	+1.91	+1.12
LiveBench: known truth, 318 questions	reported	+0.06 (0.11)	−0.01	2.12	0.94	8.1	100%	+0.06	−0.01
	bootstrap	−0.05 (0.11)	−0.03	2.22	0.93	7.9	100%	−0.05	−0.03
	iterated bootstrap	+0.08 (0.11)	+0.06	2.18	0.93	7.9	100%	+0.08	+0.06
	conditional	+0.06 (0.11)	−0.01	2.13	0.94	8.1	100%	+0.06	−0.01
	hybrid (AKM)	+0.06 (0.11)	−0.01	2.13	0.94	8.2	100%	+0.06	−0.01
	zoom interval (ZF)	+0.06 (0.11)	−0.01	2.12	0.99	12.1	100%	+0.06	−0.01
LiveBench: equal leaders, 318 questions	reported	+3.88 (0.09)	+3.70	4.30	0.76	10.2	100%	+3.96	+4.01
	bootstrap	+1.03 (0.11)	+0.92	2.37	0.94	8.5	100%	+1.14	+0.93
	iterated bootstrap	+0.46 (0.12)	+0.27	2.35	0.95	8.5	100%	+0.59	+0.31
	conditional	−0.25 (0.24)	+0.62	4.54	0.97	24.1	87%	−0.25	+0.62
	hybrid (AKM)	−0.02 (0.23)	+0.62	4.22	0.97	14.6	87%	−0.02	+0.62
	zoom interval (ZF)	+3.88 (0.09)	+3.70	4.30	0.98	16.8	100%	+3.96	+4.01
LiveBench: held-out questions	reported	−0.48 (0.21)	−0.66	4.22	–	11.5	100%	−0.48	−0.66
	bootstrap	−0.97 (0.22)	−1.06	4.58	–	11.5	100%	−0.97	−1.06
	iterated bootstrap	−0.62 (0.23)	−0.70	4.54	–	11.5	100%	−0.62	−0.70
	conditional	−0.50 (0.21)	−0.67	4.25	–	11.6	100%	−0.50	−0.67
	hybrid (AKM)	−0.50 (0.21)	−0.67	4.25	–	11.8	100%	−0.50	−0.67
	zoom interval (ZF)	−0.48 (0.21)	−0.66	4.22	–	18.0	100%	−0.48	−0.66

Table S5: Validation on Terminal-Bench 2 (89 tasks, per-task rates from repeated trials; equal-leaders outcomes simulated with five trials) and on LiveBench (318 questions, per-question credit in $[0, 1]$; equal-leaders outcomes simulated as pass or fail; 400 repetitions). Columns as in Table S4.

4 Record ladders

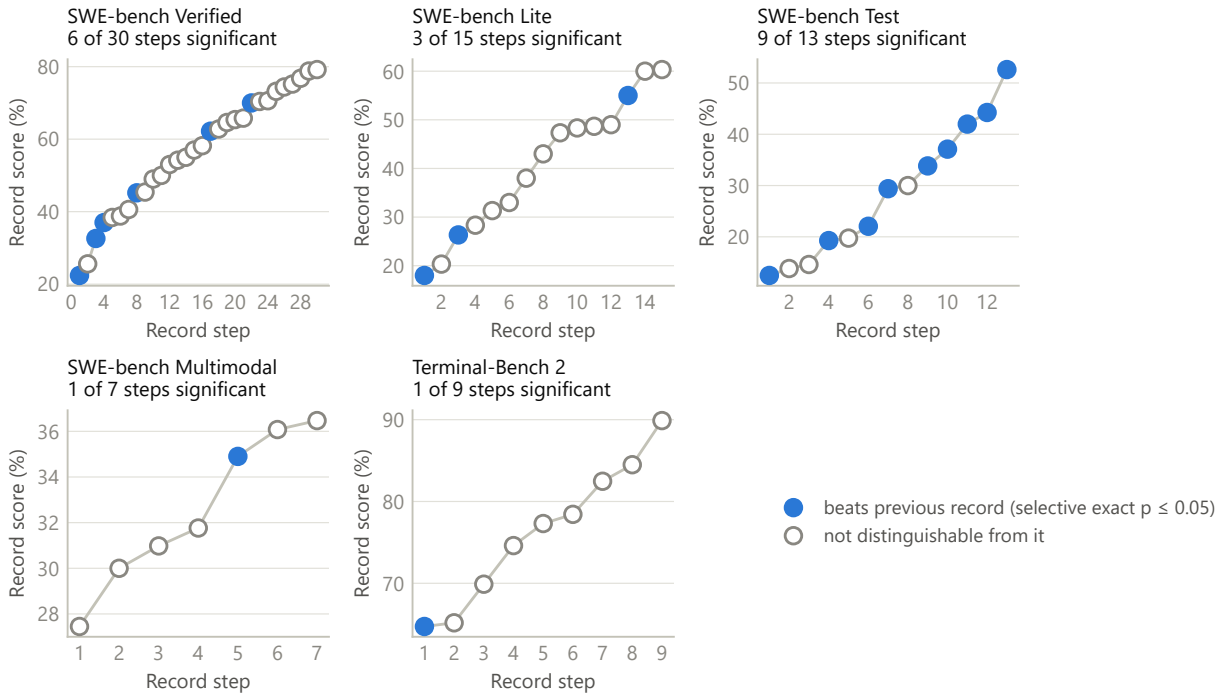


Figure S1: The record ladders of five leaderboards. A filled point is clearly better than the previous record (selective exact paired test at the 5% level); a hollow point is not.